

Riemann Sum Function

A Riemann sum is an estimation of the area under a curve using rectangles. The **right Riemann sum** R_n of a function f on the interval $[a, b]$ with n rectangles is calculated by

$$R_n = f(a + \Delta x)\Delta x + f(a + 2\Delta x)\Delta x + \cdots + f(a + n\Delta x)\Delta x = \sum_{k=1}^n f(a + k\Delta x)\Delta x,$$

where $\Delta x = \frac{b-a}{n}$, is the width of the rectangles. Write a Python function to take in a mathematical function f , two interval endpoints a and b , and the number of rectangles n , and then outputs the right Riemann sum R_n .

Using the Right Riemann Sum to Find the Area Under a Curve

Let $f(x) = \sqrt{4 - x^2}$.

- (a) Sketch a graph of f and the rectangles that make up the right Riemann sum R_4 on the interval $[0, 2]$. This can be done by hand.
- (b) Use the function you created in the previous problem to calculate R_4 .
- (c) The area below the curve of a continuous function f on the interval $[a, b]$ can be found by calculating what is called the definite integral of f over $[a, b]$. The definite integral $\int_a^b f(x) dx$ can be calculated with the following limit

$$\int_a^b f(x) dx = \text{Area Under the Curve} = \lim_{n \rightarrow \infty} R_n.$$

Calculate appropriate right Riemann sums to find the area under the curve $y = f(x)$ on the interval $[0, 2]$. Does the value of this area have a special name?